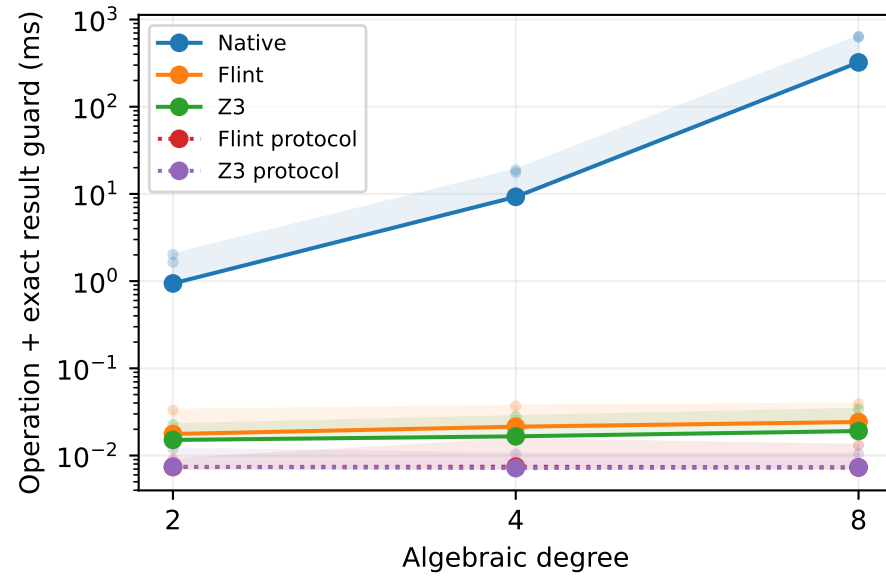
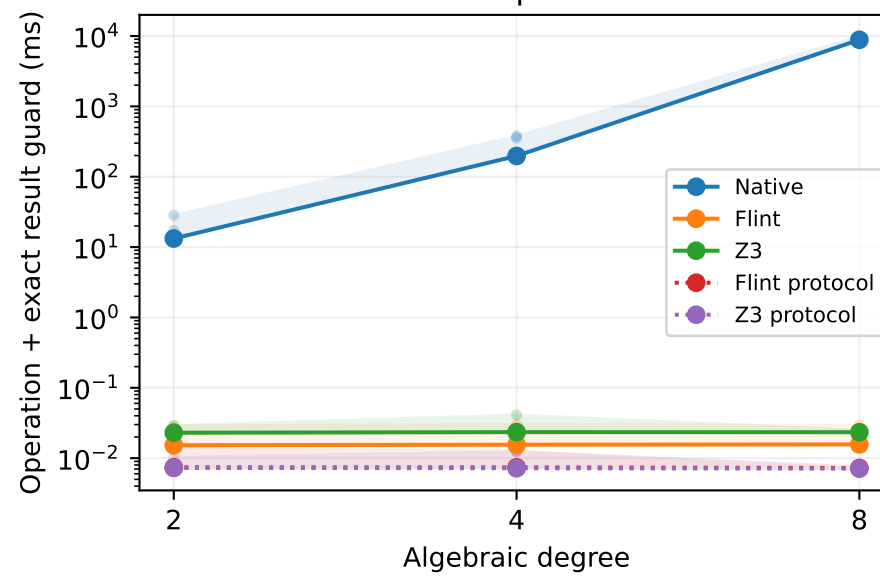


# Shipped real-algebraic scalar APIs vs FLINT qqbar and Z3 RCF

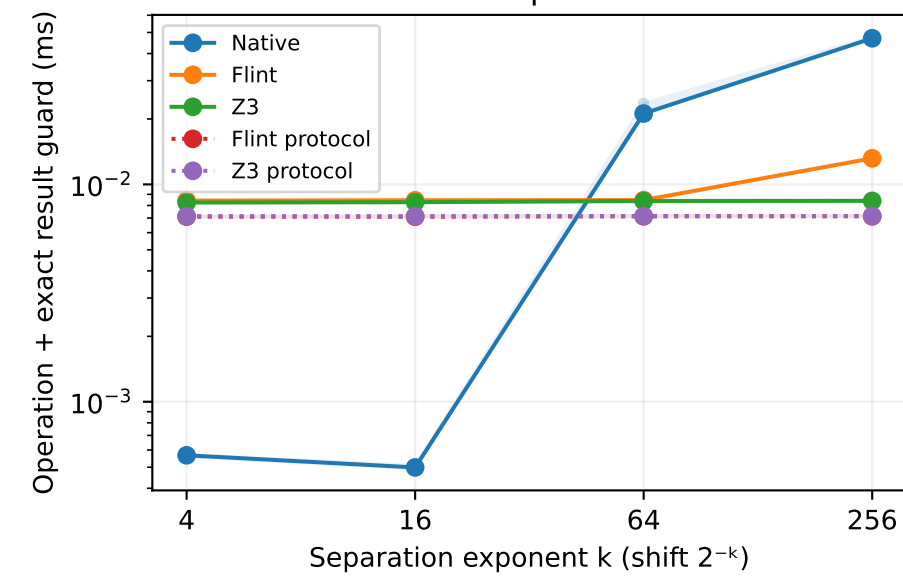
## Add



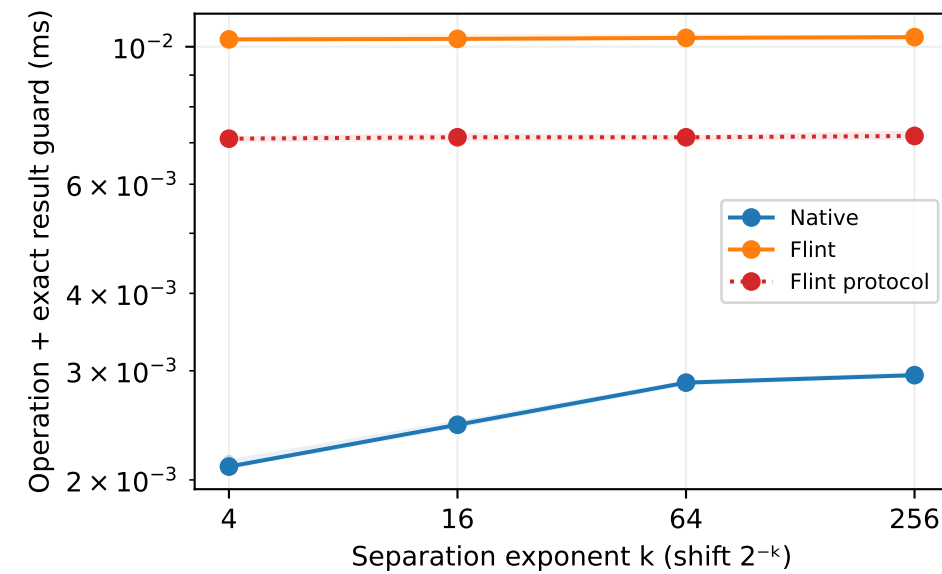
## Sqrt



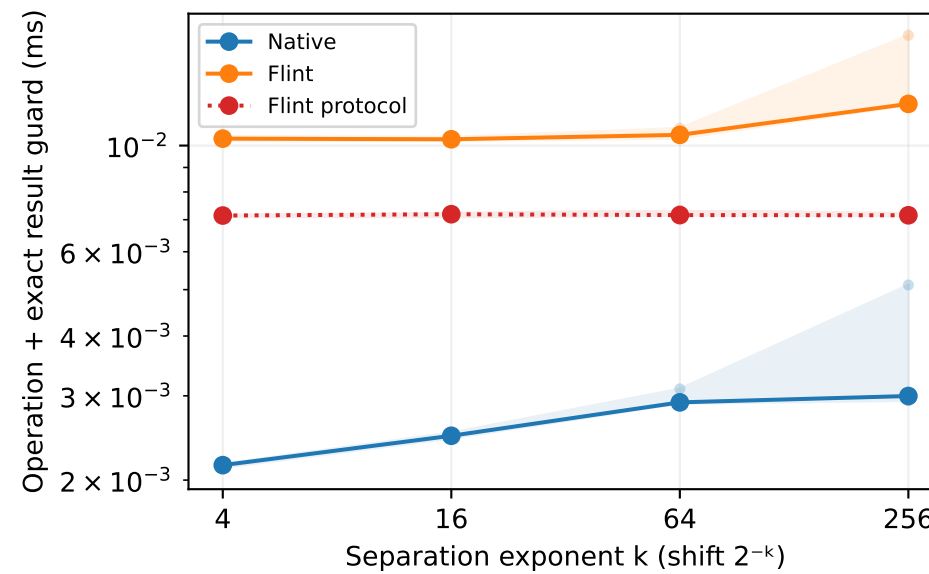
## Compare



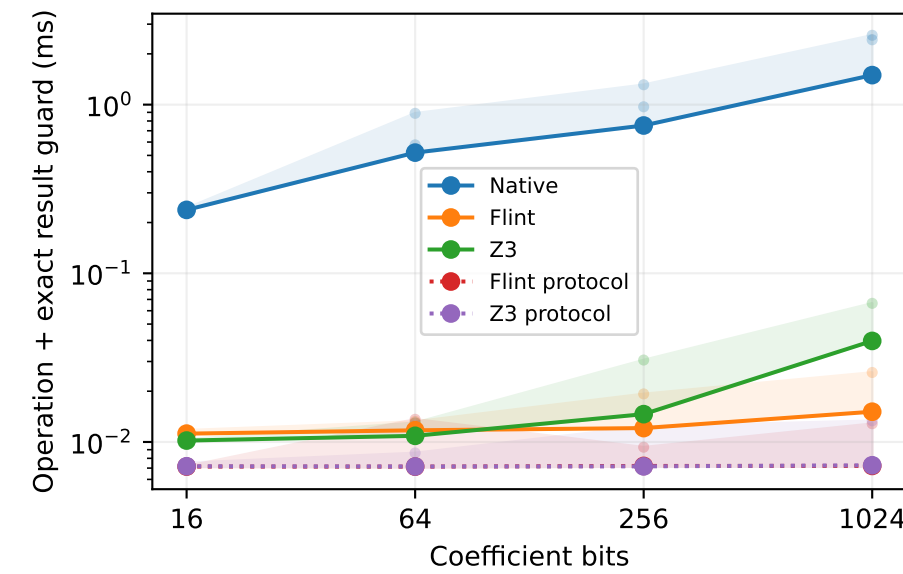
## Floor



## Ceil



## Rational



Prepared operands; native arithmetic checks canonical polynomial/sign and external arithmetic checks exact annihilation/sign.  
 No expected algebraic root is prepared. External JSON/cleanup remain timed; protocol curves are separate, without subtraction.  
 Missing Z3 floor/ceil arms denote an unavailable matching API. Whole-child caps include setup; see retained failures.  
 Source a3ddc9473f, chungus2, leased CPU 45. All completed arms and failures retained; no load filtering or rerun.